CSE221 Quiz 01 MAKEUP
Time 30 Minutes

Name: ***ID:***

Q1. Analysis the Time Complexity of this code snippet and write the upper bound time complexity: [3 x 4 = 12]

```
for i in range(1, n):
    j = 1
    while j < i * i:
        j = j - 1
```

```
for (int i = 1; i <= n; i++) {
    for (int j = 1; j * j * j <= i; j++) {
        System.out.println(j);
    }
}
```

```
int K = 0;
for (int x : arr) {  //variable x iterating through the array element
    K += x;
}
for (int i = 0; i < arr.length; i++) {

    int value = K;
            for (int j = 1; j< k-i ; j*=2)
                    {
                            print(hello)
                    }
    }
}
```

```
for (int i = 1; i <= n; i++) {
    for (int j = 1; j * j <= n; j++) {
        System.out.println(j);
    }
}
```

Q2. **T(n) = n!** [N Factorial]. What will be the upper bound time complexity of this function? [2]

Q3. You have a Binary Sequence that follows a strict rule. One or more Zeroes are followed by one or more 1s. But no 0 can be given after 1 has been set.

Example: 00001111, 01111 , 01, 00000001 etc

You have to find out where the first occurrence of “1” starts from in O(log(n)) Time complexity. [6]

CSE221 Quiz 01
Time 30 Minutes

Name: ***ID:***

Q1. Analysis the Time Complexity of this code snippet and write the upper bound time complexity: [3 x 4 = 12]

```
for (int i = 1 ; i < = n ; i++)
        {
        for (int j = 1; j < = i*i ; j++)
                        {
                                Print("hello")                        }
        }
```

```
for (int i = 1; i <= n; i++) {
   for (int j = 1; j * j * j <= i; j++) {
      System.out.println(j);
   }
}
```

```
int K = 0;
for (int x : arr) {  //variable x iterating through the array element
   K += x;
}
for (int i = 0; i < arr.length; i++) {

   int value = K;
        for (int j = 1; j< k-i ; j*=2)
                {
                        print(hello)
                }
   }
}
```

```
for (int i = 1; i <= n; i++) {
   for (int j = 1; j * j <= n; j++) {
      System.out.println(j);
   }
}
```

Q2. **T(n) = n!** [N Factorial]. What will be the upper bound time complexity of this function? [2]

Q3. You are given a sequence of sorted numbers where there are duplicate values in between.
Your task is to find the total number of duplicate values of a given integer in **O(log n)** time complexity. Write pseudocode [6]

CSE221 Quiz 02
Time 30 minutes

Name: **ID:**

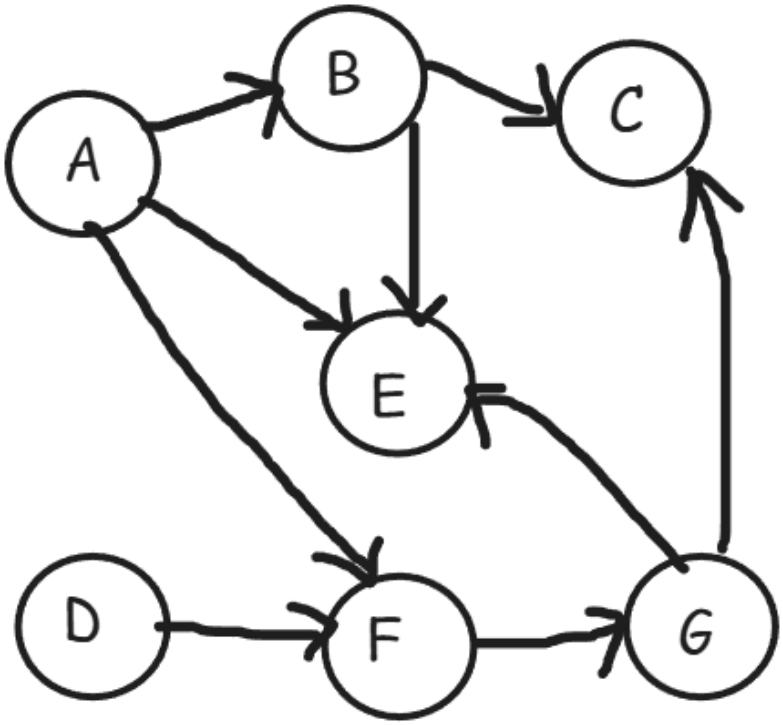
Q1. **[5]**

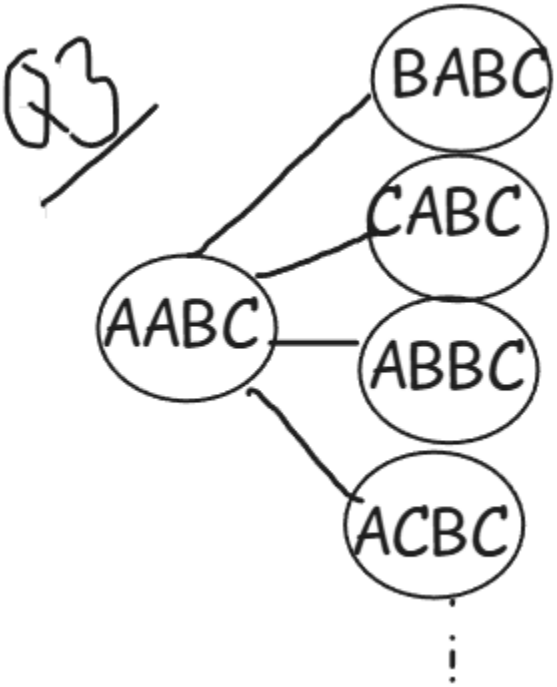
a. What is the maximum number of edges in a simple undirected graph with n vertices?
b. What is the maximum number of edges in a simple directed graph with n vertices?
c. What's the space complexity for Adjacent Matrix and Adjacent list?
d. Can BFS still explore all the nodes that it should if there is a cycle in a graph?
e. Can BFS find shortest paths for both weighted and unweighted graph?

Q2. **[10]**

Simulate BFS of this graph, make sure you are using parent, visited and level array while simulating. The source is A

.





Q3. **[5]**

You are given a DNA Sequence that has a total of 4 genes[characters in the genome]. The genes can only be either A , B or C. Now you are given an initial Genome sequence. and with each generation you can mutate the gene but you can only alter one gene at a time. You have to find how to get to the target Gene in a shortest number of mutations possible, print the decisions needed to go there from source.

Starting : A A B C

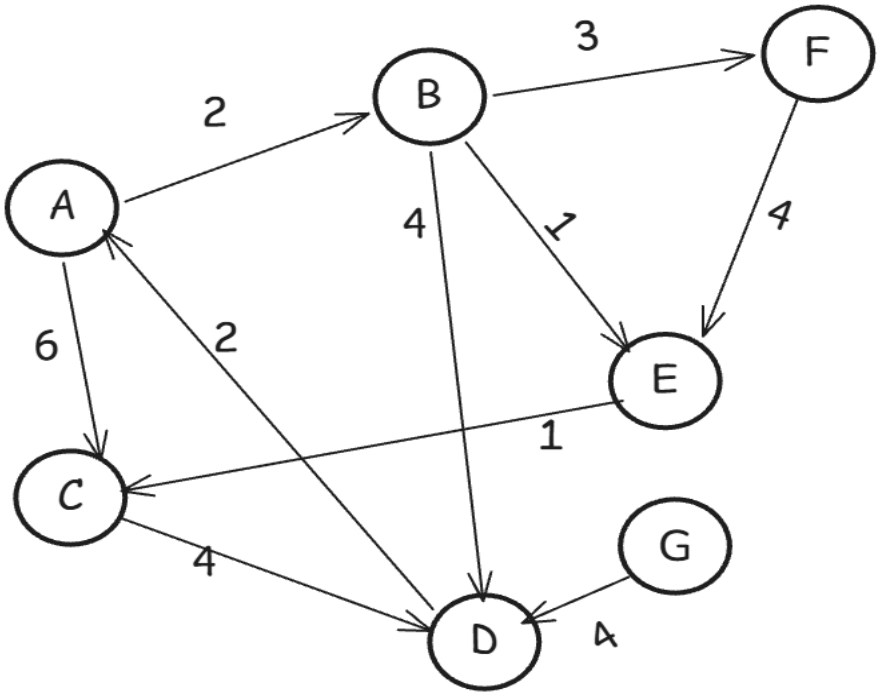
Target : A B D A

Hint is given on the right side figure. Basically you have to find all possible nodes by changing each gene with the variations it can have to gradually reach the target node. You don't need to explore all paths, just the one that will go to the solution, with how its parent tracking will be stored will be enough.

CSE221 Quiz 03
Time 30 Minutes

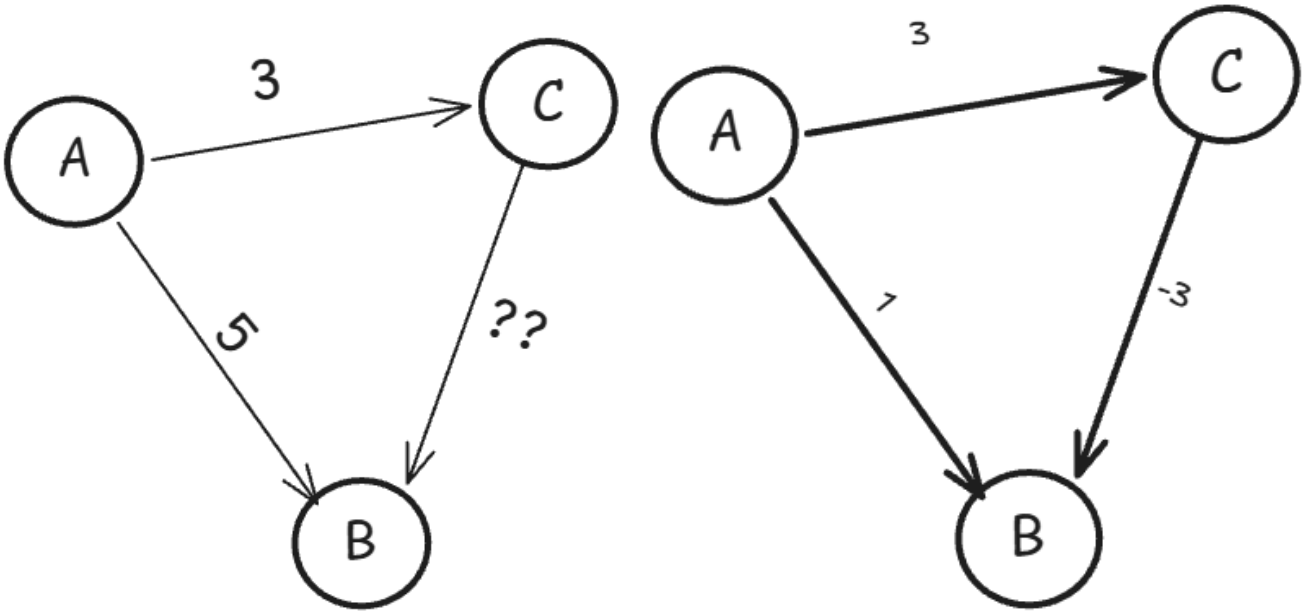
Name: ID: .

Q1. Given this graph, Simulate Dijkstra algorithm with source D while tracking the Distance Tracker, Parent Tracker, and other relevant information. After simulation, give the shortest path distance and path taken to reach these 4 nodes: C, B, D, G. [10 + 4]



Q2. a. Given the left graph, run dijkstra from source A. What will be the BC edge weight for the shortest path from A to B that has to go through C? [3]

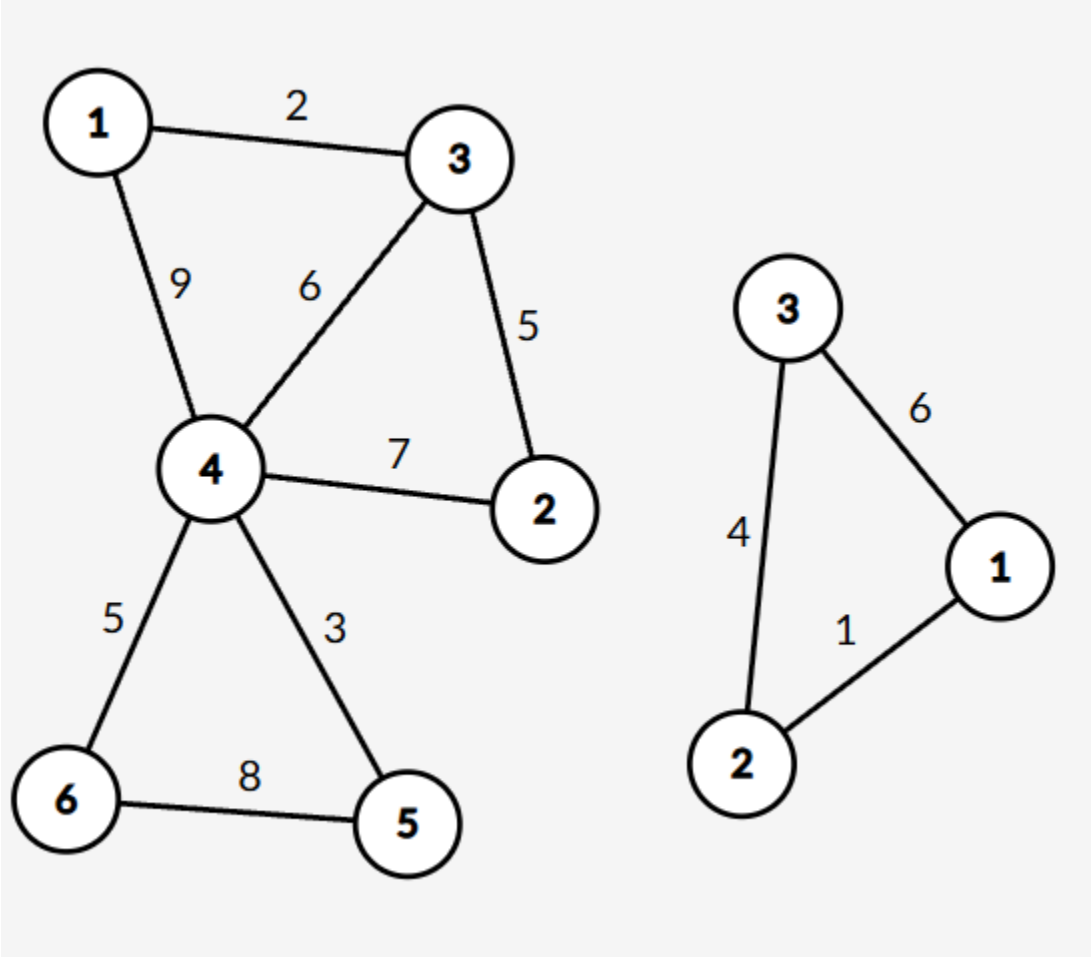
b. On the right graph. Will Dijkstra Run properly [source A]? Explain with simulation/statements. [3]



CSE221 Quiz 04
Time 25 minutes

Name: ID:

Q1. Consider this graph:



a. Run Kruskal in this graph and show the result tree: follow the necessary steps for simulation for full marks. [8]

b. Did the algorithm achieve MST? If not, then what can we add to make it an MST? . [2]

c. suppose the graph has negative weights, how does the behavior of the algorithm change? You can show a small 4 vertices graph to demonstrate if you want [3]

d. Greedy approach suggests that kruskal will always choose the smallest weight edge. Consider this:

i. A,B : 1 Chosen, ii. A,C: 3 chosen, iii. B,C: 3 Not chosen, iv. A,D:13 chosen.

Why is the bigger edge chosen before the smaller one? [3]

e. If a graph has total V vertices, and your MST has total V edges, will the MST be a valid one? Explain [4]

f. A graph is guaranteed to have a unique MST. Is it True or False? [1]

CSE221 Quiz 05 / MAKEUP
Time 30 minutes

Name: ID:

Q1. You are trying to do odd jobs to maximize your earning. You have lots of jobs in your hand, The total amount of work hours you can manage is 10 hours. But there is an issue, there is a specific job (in this case, Job 5) that you must do since that is a relevant experience for your future career. Now design a Dynamic Programming Solution to determine which jobs to take so that you can maximize your earning potential.

Job No	Duration (Hour)	Salary (thousand)
1	2	8
2	4	11
3	5	14
4	2	5
5	3	7
6	1	3

a. Analyse and draw the DP table [12]

b. Backtrack to show the optimal solution subset [5]

c. define time complexity of the problem [1]

d. In a situation for DP[x,y] where If an item's weight exceeds the current capacity, x being the current item and y is the current capacity, what will be the solution for this cell? [2]